

Software Requirements Specification (SRS)

GTIDS Online Agreement Management & Digital Signing Portal

Organization: Gramtarang Inclusive Development Services (GTIDS)

Version: 1.0

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Document Control

Item	Value
Document Name	Software Requirements Specification
System	GTIDS Online Agreement Management & Digital Signing Portal
Organization	Gramtarang Inclusive Development Services (GTIDS)
Version	1.0
Primary Workflow	Agent Sign → Employee Approve → MD Sign → Email to all three
Identity/Signing Integration	Third-party Aadhaar OTP/eSign provider
Optional Integration	DigiLocker

1. Purpose

The GTIDS Online Agreement Management & Digital Signing Portal will digitize the complete agreement lifecycle from initiation and stamping through sequential signing, approval, finalization, notification and verification. GTIDS will own and operate the application, database, document management, workflow, audit trail and reporting. A third-party regulated service will be used only for Aadhaar OTP/eSign functions as required.

2. Scope

- Create and manage agreement types and templates.
- Upload and manage physical ■100 stamp paper scans.
- Support a future e-Stamp integration without changing the core agreement engine.
- Capture agreement party and employee information.
- Verify identity using the configured third-party Aadhaar OTP service.
- Optionally retrieve or verify documents through DigiLocker.
- Generate a final PDF agreement from approved templates and data.
- Execute the mandatory sequence: Agent signs → Employee approves → MD signs.
- Maintain document versions, hashes, signatures and complete audit trails.
- Send the completed agreement by email to the Agent, Employee and MD.
- Provide QR/public verification of completed agreements.
- Provide administrative dashboards, reports and operational monitoring.

3. Business Workflow

1. Agent initiates the agreement and enters or verifies required details.

2. System validates mandatory data and attaches/allocates the applicable stamp paper.
3. Agreement PDF is generated from the selected template.
4. Agent reviews the document and signs through the configured eSign/Aadhaar OTP process.
5. After successful Agent signature, the agreement moves to Employee Approval.
6. Employee reviews the complete agreement and approves it.
7. After Employee approval, the agreement moves to MD Signature.
8. MD reviews the final agreement and signs through the configured eSign/Aadhaar OTP process.
9. System marks the agreement as COMPLETED.
10. System generates the final signed PDF, verification reference/QR and audit record.
11. System emails the completed agreement to Agent, Employee and MD.

4. Actors and Roles

Role	Key Responsibilities
Super Administrator	System configuration, roles, templates, integrations and global controls.
Agreement Administrator	Agreement types, templates, stamp inventory and operational configuration.
Agent	Initiates agreement, enters data, reviews and digitally signs.
Employee	Reviews the Agent-signed agreement and approves it.
MD	Performs final review and digital signature.
Auditor	Read-only access to agreements, signatures and audit trails.
Verification User	Verifies a completed agreement using agreement number/QR.

5. Functional Requirements

ID	Requirement	Description
FR-001	User Authentication	Authenticate internal users and enforce role-based permissions.
FR-002	Agreement Creation	Create agreements using approved agreement types/templates.
FR-003	Template Management	Create, version, approve and retire agreement templates.
FR-004	Party Data	Capture required Agent, Employee and organizational information.
FR-005	Physical Stamp	Upload, catalogue, allocate and mark physical 100 stamp paper as used.
FR-006	Stamp Validation	Prevent the same stamp paper from being allocated to multiple agreements.
FR-007	Identity Verification	Initiate Aadhaar OTP verification through the configured third-party provider.
FR-008	DigiLocker	Optionally retrieve/verify permitted documents through DigiLocker with consent.
FR-009	PDF Generation	Generate agreement PDF from template and verified data.
FR-010	Document Hash	Assign SHA-256 to generated/signed document versions.
FR-011	Agent Signature	Agent digitally signs before Employee approval.
FR-012	Employee Approval	Employee approves the Agent-signed agreement before MD signing.
FR-013	MD Signature	MD digitally signs after Employee approval.
FR-014	Sequential Workflow	Prevent later steps before preceding required steps complete.
FR-015	Rejection	Allow authorized approvers to reject with a mandatory reason.

ID	Requirement	Description
FR-016	Finalization	After MD signature, mark COMPLETED and lock normal edits.
FR-017	Audit Trail	Record material actions with actor, time, action, result and transaction references.
FR-018	Email Notification	On completion, email final agreement to Agent, Employee and MD.
FR-019	QR Verification	Provide unique verification URL/QR for completed agreements.
FR-020	Reports	Provide agreement, stamp, workflow, signature and completion reports.

6. Agreement State Model

DRAFT → READY_FOR_AGENT_SIGNATURE → AGENT_SIGNING → AGENT_SIGNED → PENDING_EMPLOYEE_APPROVAL → EMPLOYEE_APPROVING → EMPLOYEE_APPROVED → PENDING_MD_SIGNATURE → MD_SIGNING → COMPLETED. Exception states: REJECTED, CANCELLED, EXPIRED, SIGNATURE_FAILED.

7. Stamp Paper Requirements

Field	Description
Stamp Number	Unique stamp/certificate reference where available.
Denomination	e.g. ₹100.
State	State in which stamp is issued.
Issue/Purchase Date	Applicable date.
Vendor	Stamp vendor/source if recorded.
Scan	Original uploaded scan/PDF.
SHA-256	Hash of original uploaded document.
Status	AVAILABLE / ALLOCATED / USED / CANCELLED.

8. Digital Signing Requirements

- The portal shall not implement its own Aadhaar authentication mechanism.
- The selected third-party provider shall perform the Aadhaar OTP/eSign transaction.
- The portal shall store provider transaction IDs, status, timestamps and permitted reference data rather than OTP values.
- The signing service shall be abstracted behind an internal provider adapter so the provider can be changed later.
- The signed document shall be preserved and its integrity hash recorded.
- The portal shall not modify a digitally signed PDF in a way that invalidates an earlier signature.

9. Email Requirements

After MD signature and successful finalization, one completion notification shall be sent to all three parties.

Recipient	Email Content
Agent	Completion notice, agreement number and final signed PDF/link.
Employee	Completion notice, agreement number and final signed PDF/link.
MD	Completion confirmation and final signed PDF/link.

The email subsystem shall record delivery status, timestamp, message ID and failure reason where available.

10. Audit Requirements

- Agreement created
- Stamp uploaded/allocated
- Identity verification initiated/completed
- Agreement generated
- Agent viewed/signed
- Employee viewed/approved/rejected
- MD viewed/signed
- Agreement completed
- Final document generated
- Email sent to each recipient
- Document downloaded/verified

11. Non-Functional Requirements

Category	Requirement
Security	TLS, encryption at rest, RBAC, secure secrets, audit logging, OWASP-aligned controls.
Availability	Target 99.9% for production, subject to infrastructure and external provider availability.
Performance	Normal API operations should target <2 seconds excluding external provider and PDF-generation latency.
Scalability	Architecture shall support horizontal application scaling.
Backup	Automated database and document backups with tested restoration.
Auditability	Audit records shall be append-oriented and protected from normal user modification.
Privacy	Collect and retain only data required for the agreement process.
Maintainability	Modular application architecture with provider adapters.
Compatibility	Responsive web application supporting current Chrome, Edge, Safari and Firefox versions.

12. Security and Privacy

- Aadhaar OTP values shall not be stored in the GTIDS database.
- Sensitive identity data shall be minimized and access-controlled.
- Documents shall be stored in private storage with encryption and access policies.
- Administrative actions shall require stronger authentication and least-privilege access.
- Download links should be short-lived and access-controlled.
- Audit records shall be retained according to GTIDS policy and applicable legal requirements.

13. Acceptance Criteria

1. A valid Agent can create an agreement and generate the correct PDF.
2. Agent signature must be completed before Employee approval is enabled.
3. Employee approval must be completed before MD signing is enabled.
4. MD cannot sign a rejected or cancelled agreement.
5. After MD signature, the agreement becomes COMPLETED and normal content editing is disabled.

6. Final signed document is retrievable from the authorized portal.
7. Email notification is sent to Agent, Employee and MD.
8. All workflow events appear in the audit trail.
9. The QR/verification reference returns the correct agreement status without exposing sensitive data.